Report for ForestQuery into Global Deforestation, 1990 to 2016

ForestQuery is on a mission to combat deforestation around the world and to raise awareness about this topic and its impact on the environment. The data analysis team at ForestQuery has obtained data from the World Bank that includes forest area and total land area by country and year from 1990 to 2016, as well as a table of countries and the regions to which they belong.

The data analysis team has used SQL to bring these tables together and to query them in an effort to find areas of concern as well as areas that present an opportunity to learn from successes.

1. **GLOBAL SITUATION**

According to the World Bank, the total forest area of the world was 41,282,694.9 sq. km in 1990. As of 2016, the most recent year for which data was available, that number had fallen to 39,958,245.9 sq. km., a loss of 1324449 sq. km., or -3.21%.

The forest area lost over this time period is slightly more than the entire land area of Peru listed for the year 2016 (which is 1,280,000 sq. km).

2. **REGIONAL OUTLOOK**

In 2016, the percent of the total land area of the world designated as forest was 31.34%. The region with the highest relative forestation was Latin America & Caribbean, with 46.16%, and the region with the lowest relative forestation was the Middle East & North Africa, with 2.07% forestation.
In 1990, the percent of the total land area of the world designated as forest was 32.21%. The region with the highest relative forestation was Latin America & Caribbean, with 51.03%, and the region with the lowest relative forestation was the Middle East & Africa, with 1.78% forestation.

Table 2.1: Percent Forest Area by Region, 1990 & 2016:

Region	1990 Forest Percentage	2016 Forest Percentage
Latin America & Caribbean	51.03	46.16
Europe & Central Asia	37.28	38.04
North America	35.65	36.04
Sub-Saharan Africa	30.67	28.79
East Asia & Pacific	25.78	26.36
South Asia	16.51	17.51
Middle East & North Africa	1.78	2.07
World	32.42	31.38

The only regions of the world that decreased in percent forest area from 1990 to 2016 were Latin America & Caribbean (dropped from 51.03% to 46.16%) and Sub-Saharan Africa (30.67% to 28.79%). All other regions actually increased in forest area over this time period. However, the drop in forest area in the two aforementioned regions was so large, the percent forest area of the world decreased over this time period from 32.42% to 31.38%.

3. **COUNTRY-LEVEL DETAIL**

A. SUCCESS STORIES

There is one particularly bright spot in the data at the country level, China. This country actually increased in forest area from 1990 to 2016 by 527,229.06 sq. km. It would be interesting to study what has changed in this country over this time to drive this figure in the data higher. The country with the next largest increase in forest area from 1990 to 2016 was the United States of America, but it only saw an increase of 79,200 sq. km., much lower than the figure for China.

China and the United States of America are of course very large countries in total land area, so when we look at the largest percent change in forest area from 1990 to 2016, we aren't surprised to find a much smaller country listed at the top. Iceland increased in forest area by 213.66% from 1990 to 2016.

B. LARGEST CONCERNS

Which countries are seeing deforestation to the largest degree? We can answer this question in two ways. First, we can look at the absolute square kilometer decrease in forest area from 1990 to 2016. The following 3 countries had the largest decrease in forest area over the time period under consideration:

Table 3.1: Top 5 Amount Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Absolute Forest Area Change
Brazil	Latin America & Caribbean	541,510 sq. km.
Indonesia	East Asia & Pacific	282,194 sq. km.
Myanmar	East Asia & Pacific	107,234 sq. km.
Nigeria	Sub-Saharan Africa	106,506 sq. km.
Tanzania	Sub-Saharan Africa	102,320 sq. km.

The second way to consider which countries are of concern is to analyze the data by percent decrease.

Table 3.2: Top 5 Percent Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Pct Forest Area Change
Togo	Sub-Saharan Africa	75.45%
Nigeria	Sub-Saharan Africa	61.80%
Uganda	Sub-Saharan Africa	59.13%
Mauritania	Sub-Saharan Africa	46.75%
Honduras	Latin America & Caribbean	45.03%

When we consider countries that decreased in forest area percentage the most between 1990 and 2016, we find that four of the top 5 countries on the list are in the region of Sub-Saharan Africa. The countries are Togo, Nigeria, Uganda, and Mauritania. The 5th country on the list is Honduras, which is in the Latin America & Caribbean region.

From the above analysis, we see that Nigeria is the only country that ranks in the top 5 both in terms of absolute square kilometer decrease in forest as well as percent decrease in forest area from 1990 to 2016. Therefore, this country has a significant opportunity ahead to stop the decline and hopefully spearhead remedial efforts.

C. QUARTILES

Table 3.3: Count of Countries Grouped by Forestation Percent Quartiles, 2016:

Quartile	Number of Countries
0 - 25%	85
25 - 50%	72
50 - 75%	38
75 - 100%	9

The largest number of countries in 2016 were found in the 0-25% quartile.

There were 9 countries in the top quartile in 2016. These are countries with a very high percentage of their land area designated as forest. The following is a list of countries and their respective forest land, denoted as a percentage.

Table 3.4: Top Quartile Countries, 2016:

Country	Region	Pct Designated as Forest
Suriname	Latin America & Caribbean	98.26%
Micronesia, Fed. Sts.	East Asia & Pacific	91.86%
Gabon	Sub-Saharan Africa	90.04%
Seychelles	Sub-Saharan Africa	88.41%
Palau	East Asia & Pacific	87.61%
American Samoa	East Asia & Pacific	87.50%
Guyana	Latin America & Caribbean	83.90%

Lao PDR	East Asia & Pacific	82.11%
Solomon Islands	East Asia & Pacific	77.86%

4. RECOMMENDATIONS

Write out a set of recommendations as an analyst on the ForestQuery team.

- *What have you learned from the World Bank data?*
- *Which countries should we focus on over others?*

Learnings of World Bank data:

This report relies solely on figures published by the World Bank and does not factor in related influences like changes in population, economic conditions, or ongoing domestic or international conflicts. Each of these could meaningfully affect whether forest land expands or shrinks.

The countries that should be focused upon:

Attention should be directed toward the countries that have experienced the steepest declines in forest land, whether measured in absolute square kilometers or as a percentage of total forest cover. This group includes Brazil, Indonesia, Myanmar, Nigeria, Tanzania, St. Martin, Togo, Uganda, and Mauritania. Understanding what drove these losses between 1990 and 2016 will be key to curbing further decline and, where feasible, restoring forest land. Many of the hardest-hit areas have also experienced conflict, yet they remain rich in biodiversity and natural resources worth safeguarding.

5. APPENDIX: SQL Queries Used

Create View

```
-- Create a view called "forestation"
-- Joins forest_area, land_area, and regions tables
DROP VIEW IF EXISTS forestation;
CREATE VIEW forestation
AS SELECT
fa.country_code,
fa.country_name,
fa.year,
fa.forest_area_sqkm,
la.total_area_sq_mi * 2.59 AS total_area_sqkm,
-- Calculate the percent of total land that is forest area
(fa.forest_area_sqkm/(la.total_area_sq_mi * 2.59) * 100) AS
perc_land_is_forest,
r.region,
```

```
r.income_group
FROM forest_area fa
JOIN land_area la
ON fa.country_code = la.country_code
AND fa.year = la.year
JOIN regions r
ON la.country_code = r.country_code;
```

Global Situation

```
/* a. What was the total forest area (in sq km) of the world in 1990? */
SELECT country_name, forest_area_sqkm, year
FROM forestation
WHERE country_name = 'World' AND year = 1990;

/* b. What was the total forest area (in sq km) of the world in 2016? */
SELECT country_name, forest_area_sqkm, year
FROM forestation
WHERE country_name = 'World' AND year = 2016;

/* c. What was the change (in sq km) in the forest area of the world
   from 1990 to 2016? */
SELECT (f1.forest_area_sqkm - f2.forest_area_sqkm) AS forest_area_change_sq_km
FROM forestation AS f1, forestation AS f2
WHERE f1.year = '2016' AND f1.country_name = 'World'
AND f2.year = '1990' AND f2.country_name = 'World';

/* d. What was the percent change in forest area of the world between
   1990 and 2016? */
SELECT
(f1.forest_area_sqkm - f2.forest_area_sqkm) AS forest_area_change_sq_km,
((f1.forest_area_sqkm - f2.forest_area_sqkm) / f2.forest_area_sqkm) * 100
AS forest_area_percent_change
FROM forestation AS f1, forestation AS f2
WHERE f1.year = '2016' AND f1.country_name = 'World'
AND f2.year = '1990' AND f2.country_name = 'World';

/* e. If you compare the amount of forest area lost between 1990 and
   2016, to which country's total area in 2016 is it closest to? */
SELECT country_name, year, total_area_sqkm
FROM forestation
WHERE year = 2016
AND (total_area_sqkm) < ((
    SELECT forest_area_sqkm FROM forestation
    WHERE country_name = 'World' AND year = 1990) -
    (SELECT forest_area_sqkm FROM forestation
    WHERE country_name = 'World' AND year = 2016))
ORDER BY total_area_sqkm DESC
LIMIT 1;
```

Regional Outlook

```
/* a. Percent forest of the entire world in 2016; highest and lowest
   region, to 2 decimal places. */
SELECT ROUND(CAST(SUM(forest_area_sqkm) / SUM(total_area_sqkm) * 100
AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 2016;

SELECT region, ROUND(CAST(SUM(forest_area_sqkm) /
NULLIF(SUM(total_area_sqkm), 0) * 100 AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 2016
GROUP BY 1
ORDER BY 2 DESC;

SELECT region, ROUND(CAST(SUM(forest_area_sqkm) /
NULLIF(SUM(total_area_sqkm), 0) * 100 AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 2016
GROUP BY 1
ORDER BY 2;

/* b. Percent forest of the entire world in 1990; highest and lowest
   region, to 2 decimal places. */
SELECT SUM(forest_area_sqkm) / SUM(total_area_sqkm) * 100 AS forest_area_world
FROM forestation
WHERE year = 1990;

SELECT region, ROUND(CAST(SUM(forest_area_sqkm) /
NULLIF(SUM(total_area_sqkm), 0) * 100 AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 1990
GROUP BY 1
ORDER BY 2 DESC;

SELECT region, ROUND(CAST(SUM(forest_area_sqkm) /
NULLIF(SUM(total_area_sqkm), 0) * 100 AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 1990
GROUP BY 1
ORDER BY 2;

/* c. Which regions of the world DECREASED in forest area from 1990
   to 2016? */
SELECT
region,
SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS forest_area_1990,
SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) AS forest_area_2016
FROM forestation
GROUP BY region
HAVING SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) <
```

```
SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END);
```

Country-level Detail

```
/* a. Which 5 countries saw the largest amount decrease in forest area
   from 1990 to 2016? */
SELECT
country_code, country_name,
SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS forest_area_1990,
SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) AS forest_area_2016,
(SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) -
 SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)) AS forest_area_change
FROM forestation
WHERE country_name <> 'World'
GROUP BY country_code, country_name
ORDER BY forest_area_change ASC
LIMIT 6;

/* b. Which 5 countries saw the largest percent decrease in forest
   area from 1990 to 2016? */
SELECT
country_code, country_name,
SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS forest_area_1990,
SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) AS forest_area_2016,
ROUND(CAST((SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) -
SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)) /
SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) * 100
AS NUMERIC), 2) AS percent_change
FROM forestation
GROUP BY country_code, country_name
HAVING SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) > 0
ORDER BY percent_change ASC
LIMIT 6;

/* c. If countries were grouped by percent forestation in quartiles,
   which group had the most countries in it in 2016? */
SELECT
DISTINCT(quartiles),
COUNT(country_name) OVER (PARTITION BY quartiles) AS countries
FROM (SELECT country_name,
CASE WHEN perc_land_is_forest <= 25 THEN '0-25%'
WHEN perc_land_is_forest <= 75 AND perc_land_is_forest > 50 THEN '50-75%'
WHEN perc_land_is_forest <= 50 AND perc_land_is_forest > 25 THEN '25-50%'
ELSE '75-100%'
END AS quartiles
FROM forestation
WHERE (perc_land_is_forest IS NOT NULL AND year = 2016)
AND country_name <> 'World') quart;

/* d. List all of the countries that were in the 4th quartile
   (percent forest > 75%) in 2016. */
SELECT country_name, region, perc_land_is_forest
```

```
FROM forestation
WHERE year = 2016
AND country_name <> 'World'
AND perc_land_is_forest > 75
ORDER BY 3 DESC;

/* e. How many countries had a percent forestation higher than the
   United States in 2016? */
WITH us_forest_percentage AS (
    SELECT perc_land_is_forest
    FROM forestation
    WHERE country_code = 'USA' AND year = 2016
)
SELECT COUNT(*) AS countries_higher_than_us
FROM forestation
WHERE year = 2016
AND perc_land_is_forest > (SELECT perc_land_is_forest FROM us_forest_percentage);
```